

Zhengnan FU

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EDUCATION

City University of HongKong	2024/09-2028/06
Department of Electronic Engineering	
Ph.D Student of Philosophy in Electronic Engineering	
• Research Focus: Brain-computer interface analog front end, analog spike converter, analog in-memory computing.	
University of Chinese Academy of Science	2021/06-2024/06
Shanghai Institute of Microsystem and Information Technology	
Master's degree in Microelectronics and solid-state electronics	
• Research Focus: Readout circuits for infrared focal plane arrays (IRFPA) and mixed-signal IC.	
Central South University	2017/09-2021/06
School of Physics and Electronics	
Bachelor's degree in electronic information science and technology	

AWARDS

• First prize of the RoboMaster2020 Final Competition	2020/08
• First prize of the RoboMaster2020 Competition in Embedded System (ranked the second)	2020/08
• Greimey Innovation Practice Team Award In Central South University (ten winners every year in CSU)	2020/12
• China Postgraduate IC Innovation Competition Excellence Award	2023/08
• Third Prize of Electronic Design Contest of Central South University	
• Third prize in the southern division of Robomaster2019 National College Robot Competition	
• Third prize of intelligent vehicle group in Hunan Robot Competition	
• Annual Third-class Scholarship of Central South University (3 consecutive years from 2018- 2020)	

RESEARCH

- Neuromorphic Computing for BMI Systems**
Led the development of a neuromorphic computing-based Brain-Machine Interface (BMI) chip, incorporating Spiking Neural Networks (SNN) to efficiently decode and process brain signals in real-time. Focused on low-power, high-integration designs to enable multi-channel EEG signal acquisition with high fidelity. This innovative approach significantly reduces power consumption while maintaining signal resolution, contributing to advancements in neurorehabilitation, mobile healthcare, and human-computer interaction for wearable and portable devices.

- **Multi-channel Signal Acquisition and Ultra-Low Power Circuit Design**

Engineered ultra-low noise, ultra-low power circuits for multi-channel EEG signal acquisition, replacing traditional Nyquist-sampling ADC with Delta modulation-based analog pulse conversion for signal sparse coding. This design minimizes power usage and reduces system complexity. Further integrated nonlinear in-memory computation techniques, enhancing processing speed, reducing noise, and overcoming the power and latency limitations typically found in conventional digital circuits. The solution provides a scalable, high-performance architecture suitable for next-generation neurotechnology applications.

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STUDENTS ACTIVITIES

Electrical and Electronic Maker Space of Central South University

2018/10-2019/10

Manager of Technology Department

As a TA in the university-wide elective course "Theory and Practice of Electronic Design and Manufacture".

GRADUATION PROJECT

Master's Thesis: Research on High Frame Rate and Low Power Infrared Focal Plane Array Readout Circuit

Bachelor's Thesis: Design and Verification of video Monitoring based on FPGA

SKILL CERTIFICATES AND OTHERS

- **Language:** Chinese (Mother Tongue), English
- **Design and simulation:** Cadence Virtuoso IC617, Calibre, Verilog, VerilogA, Quartus II, Vivado, Altium Designer
- **Tool:** Matlab, LaTeX, Origin, Visio, Keil, Github, Java, Python, C, C++, ChatGPT
- **Hobby:** Electronic design DIY, Badminton, Swim.

Published/Accepted Journal:

Z.Fu, Z. Fu, Z. Tan and J. Cao, "A Pixel-Level, High Dynamic, Low-Power, Digital-Readout-Integrated Circuit With Hybrid Analog-to-Digital Converter for Infrared Focal-Plane Arrays," in IEEE Sensors Letters, vol. 7, no. 8, pp. 1-4, Aug. 2023, Art no. 7003604, doi: 10.1109/LSSENS.2023.3290983.

Published/Accepted Conference:

J. Yang, S. Dong, **Z. Fu**, H. Shang and A. Basu, "High Energy-efficiency and Low latency In-Memory Computing using Analog Accumulator and In-Memory ADC with shared References" 2025 62th ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, USA, 2025